FOSSDAQ

di-4x-5V

Build Manual
4-channel digital pulse / frequency input card on TS-35 DIN rail

Project:	FOSSDAQ
Module:	di-4x-5V
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1 Basic data

> *The `di-4x-5V` card measures pulse frequencies from up to **four** sensors. It uses an Arduino Nano R4 with 5 V digital logic and sends the measured frequencies to a Raspberry Pi via USB. The Raspberry Pi / Node-RED performs the sensor-specific conversion, for example from frequency to flow in l/min.*

Property	Value
Function	4 digital pulse / frequency inputs
Controller	Arduino Nano R4
Input pins	D2, D3, D8 and D12
Signal	Digital pulses between GND and 5 V
Bus supply	24 V DC; common ground
Sensor supply	Adjustable LM2596S output; maximum 21 V, subject to the sensor supply rating
Transfer	Frequency multiplied by 100; USB serial to the Raspberry Pi
Example	75.56 Hz is transmitted as 7556
Mounting	Printed enclosure for a TS-35 DIN rail
Isolation	No galvanic isolation between the sensors and the controller; common ground

Signal compatibility

Only connect signals compatible with the Nano R4′s 5 V digital inputs. A sensor supply rating alone does not establish signal compatibility. Do not connect 24 V signals directly to the input pins.

The DC/DC converter supplies the sensors and the Arduino Nano R4′s VIN pin. The Nano R4 accepts a VIN supply of **up to 21 V**. If a sensor requires more than 5 V, the converter output can be adjusted accordingly, to a **maximum of 21 V**, provided the selected voltage is permitted for every connected sensor. The digital signal inputs remain limited to 5 V.

2 Components

Parts required for one card (see `../bom/` for the machine-readable bill of materials):

Qty	Part	Order code	Supplier	Purpose
1	Arduino Nano R4	ARD NANO R4	Reichelt	Pulse acquisition and USB serial communication
2	Double-sided perfboard, 80 × 20 mm, 2.54 mm pitch	LR-DS-28	Reichelt	Sensor terminal carrier and a cut strip for the bus terminal
4	6-pole spring terminal AST 025-06	AST 025-06	Reichelt	One sensor channel per terminal; three poles used per terminal in the schematic
1	4-pole spring terminal AST 025-04	AST 025-04	Reichelt	24 V / GND input and pass-through
1	LM2596S DC/DC module with display	LM2596S	Reichelt	Local sensor & Nano R4 supply
1	USB-A data cable with USB-C connector for the Nano R4	-	free choice	Connection to the Raspberry Pi
≈ 4 m	stranded wire H05V-K 1 x 0.75 mm², red / black / green / gray	-	Conrad	wire colors freely selectable; recommended as shown in the schematic
≥ 12	Screws M2.3 x 6mm & M1.2 x 6mm, self-tapping, PA Phillips Head	B0G6SWF81Z	Amazon	cover screws, thread directly into the printed plastic
-	PLA Filament, 1.75 mm	-	3D printing the enclosure and latch pin	PETG as alternative

Tools: soldering iron, solder, fine saw, side cutter, wire stripper, multimeter and suitable screwdrivers. Deburr cut perfboard edges and printed parts.

3 Circuit

3.1 Schematic

The card consists of a 24 V bus terminal, an LM2596S converter, an Arduino Nano R4 and four sensor terminals. The sensor ground and controller ground share a common reference. Editable circuit sources belong in the module's `hardware/kicad/` directory.

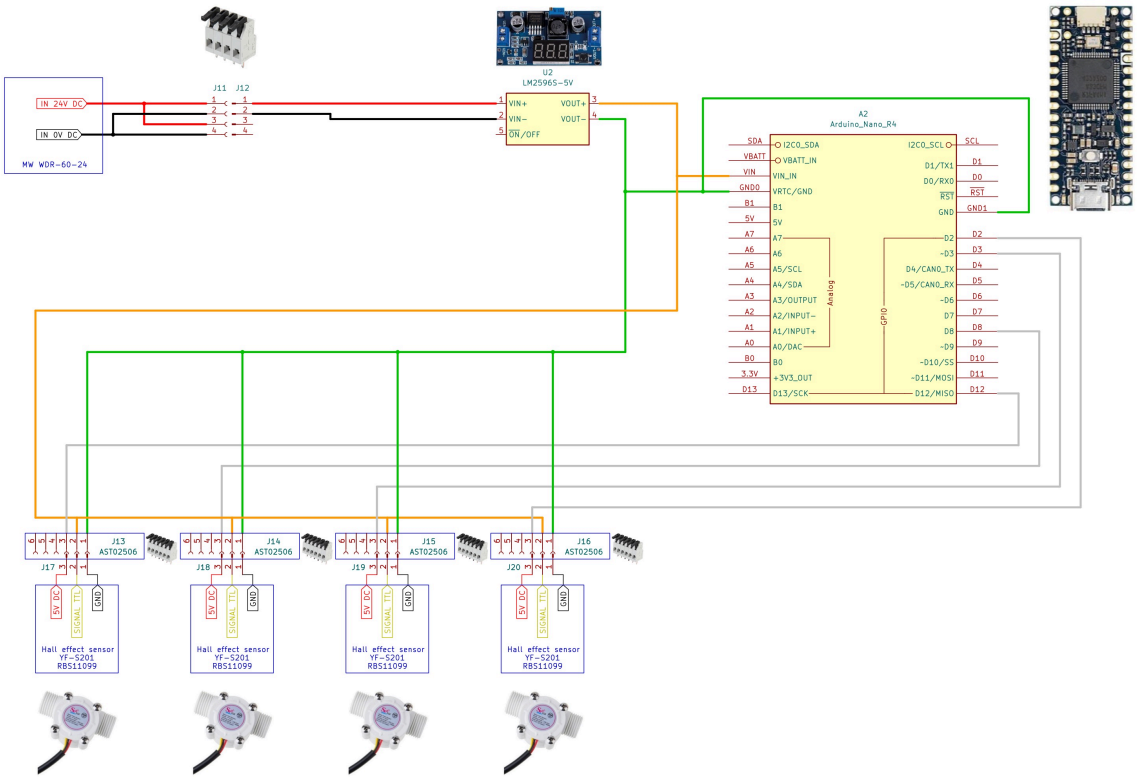


Figure 1: Schematic of the di-4x-5V card

Section	Function
Bus terminal	Receives 24 V and GND; passes both rails to the next module
DC/DC converter	Provides the common sensor and VIN supply; adjustable up to 21 V according to sensor requirements
Nano R4	Measures the four input frequencies and sends scaled values via USB
Sensor terminals	Provide supply, signal and ground connections for four sensors

3.2 Current budget

Arduino and sensors share the 5 V output of the LM2596S converter, so this module is the limiting element:

Item	Current
LM2596S, 5 V output	up to 2 A continuous
Arduino Nano R4	≈ 50 mA
4 sensors at 10-30 mA each	≈ 40-120 mA
Total	≈ 90-170 mA

The total load of roughly 90–170 mA is well below the converter's 2 A rating. Check the datasheet of each sensor for its actual supply current and ensure the total does not exceed the converter's rating. Only power-hungry sensors drawing several hundred mA each may require a separate supply.

4 Sensor connections and conversion

4.1 Terminal allocation

Each 6-pole terminal is assigned to **one** channel. The schematic uses poles 1–3; poles 4–6 have no sensor connection shown. Check the physical orientation against the schematic and the enclosure labels before connecting a sensor.

Terminal pole	Function
1	GND / 0 V
2	Sensor supply from the converter; +5 V shown in the schematic
3	Digital sensor signal
4–6	Unused in the supplied schematic

The following connections are taken from the schematic. The order of channels in the transmitted data must match the firmware and enclosure labels.

Schematic terminal	Signal pin on Nano R4
J13	D12
J14	D8
J15	D3
J16	D2

4.2 YF-S201 flow sensor

Property	Value
Principle	Hall-effect pulse output
Connection on this card	+5 V, digital signal, GND
Nominal conversion factor	7.5 Hz per l/min
Sensor relationship	`f_Hz = 7.5 * Q_l_min`
Transmitted value	`x = 100 * f_Hz`
Pi / Node-RED conversion	`Q_l_min = x / 750`

For example, 75.56 Hz is transmitted as 7556:

$$Q = \frac{7556}{750} \approx 10.07 \text{ l/min}$$

The factor 7.5 is the nominal YF-S201 conversion used here; verify it against the exact sensor datasheet and calibrate if required. For another sensor, change its conversion on the Pi while retaining the card’s frequency scaling.

5 Enclosure and 3D printing

The enclosure contains the terminal carriers, controller and converter. The cover closes the side; a separate printed latch pin secures the card to the DIN rail. Editable CAD files belong in `hardware/cad/`; print files belong in `hardware/print/`.

Part	FreeCAD file	Function
Main body	`enclosure_body+slate.FCStd`	Holds the electronics and provides the terminal openings and DIN-rail mounting
Lid	`enclosure_lid.FCStd`	Closes the enclosure and is fastened with screws
Latch pin	`enclosure_pin.FCStd`	Locks the module onto the DIN rail

5.1 Main body

The main body holds the carrier board with the terminals, the LM2596S converter and the Arduino Nano R4. The printed part has openings for the 24 V bus terminal, the four sensor terminals and the USB-C connector. The DIN-rail latch pin is inserted into the main body and secures the module to the rail.

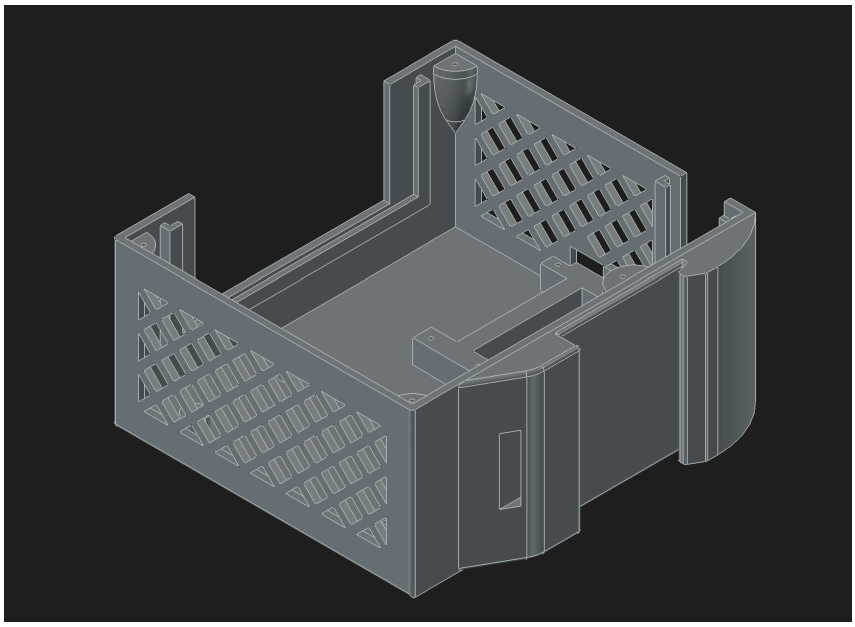


Figure 2: Open main enclosure

5.2 Lid

The lid closes the enclosure and is fastened with four self-tapping screws.

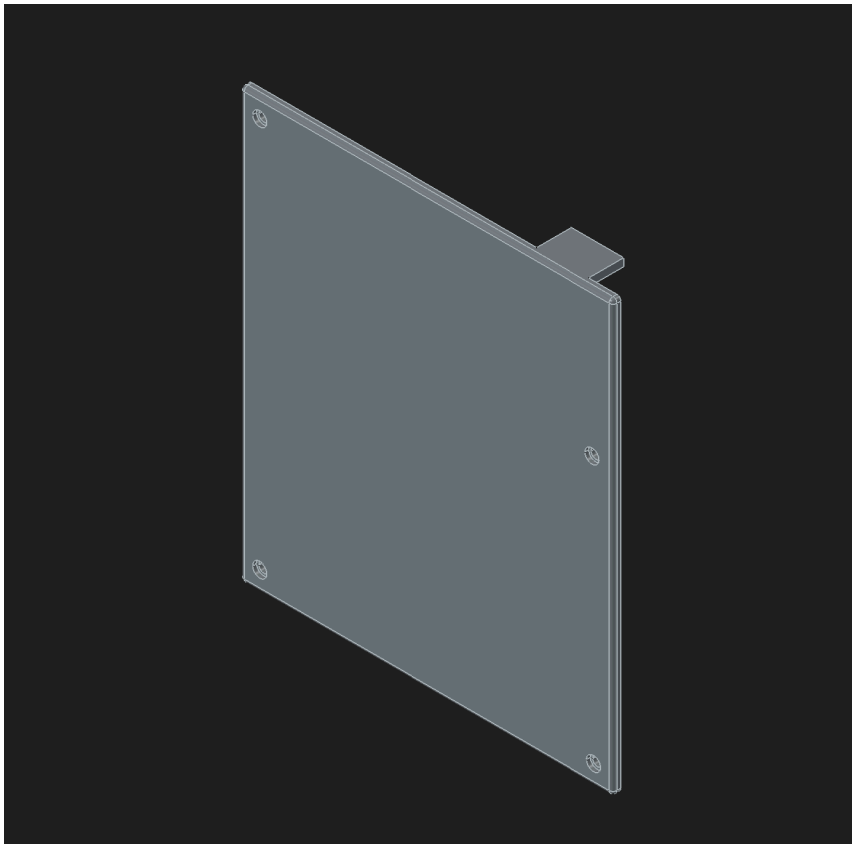


Figure 3: Side cover

5.3 DIN-rail latch pin

The pin is identical design to the pi-base pin: a separately printed springy element. Pushing the enclosure onto the rail makes it snap in; to release it, pull the pin back with a flat screwdriver or similar tool.

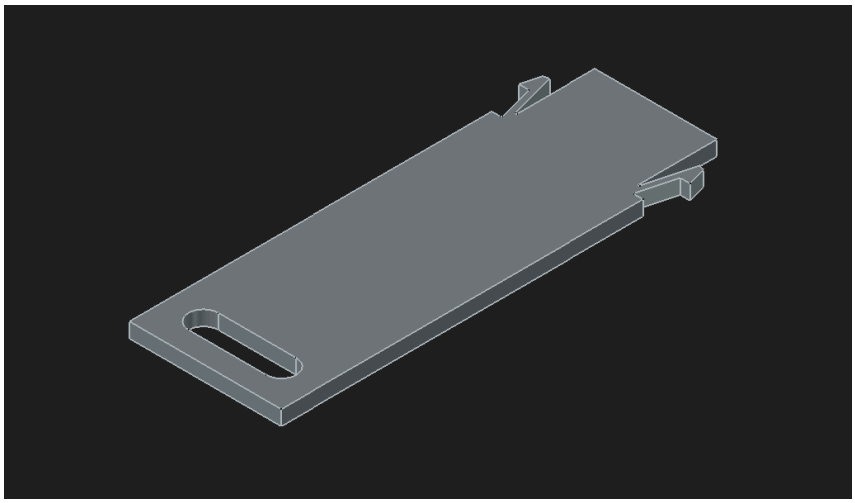


Figure 4: DIN-rail latch pin

5.4 3D printing

The print process is identical to the pi-base module.

Parameter	Value
Filament	PLA or alternatively PETG
Nozzle / bed (PLA)	210 °C / 60 °C
Nozzle / bed (PETG)	230 °C / 80 °C
Supports	not required
Attention	thin walls of the slide-in slot - check the first layer carefully

6 24 V bus terminal assembly

The bus input sits on a narrow strip cut from the 80 × 20 mm perfboard. Two terminal poles receive 24 V and GND, and the second pair provides pass-through for the next module. Bridge only matching supply potentials.

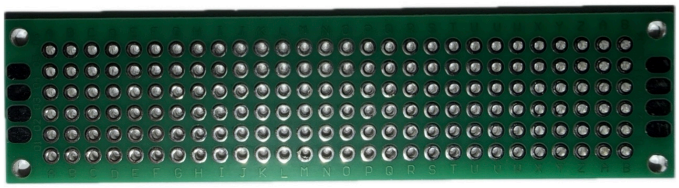


Figure 5: Raw double-sided perfboard, 80 × 20 mm.

1. **Cut and deburr the strip.** Use the marked cut lines as a guide and check the fit in the enclosure before soldering.
2. **Mount the 4-pole terminal.** Solder the AST 025-04 to the strip and connect the two 24 V poles to each other and the two ground poles to each other.
3. **Check continuity.** Verify the pass-through connections and confirm there is no short circuit between 24 V and GND.
4. **Install the carrier.** Slide the completed bus-terminal carrier into its guide.

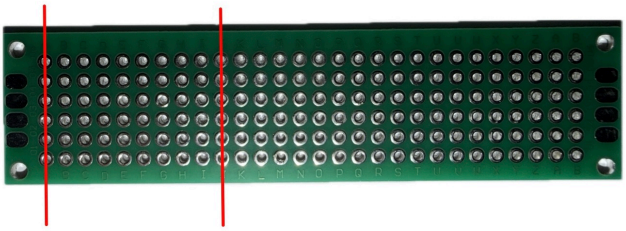


Figure 6: Marked cuts for the bus-terminal carrier.

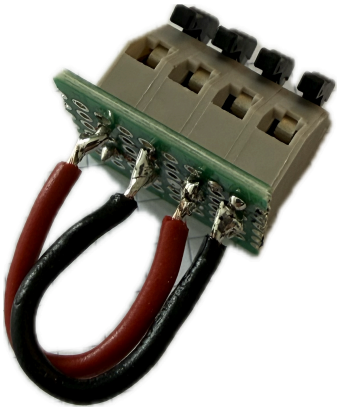


Figure 7: Bus terminal with pass-through bridges, solder side.

7 Sensor terminal carrier board

It carries the four 6-pole terminals; one terminal is assigned to each sensor channel.

> **Allow clearance for the enclosure wall**
>
> Leave approximately 2 mm between the terminal bodies and the carrier board where the enclosure design requires it. Test the fit before completing the solder joints: the enclosure wall must fit into this gap.

1. **Place the four terminals.** Maintain the required orientation and spacing.
2. **Solder from the rear.** Check that adjacent signal poles remain separate.
3. **Bridge the supply rails.** Join the four sensor-supply connections and the four GND connections. Follow the schematic rather than assuming wire colors.
4. **Prepare four signal wires.** Each sensor signal must have its own wire to the assigned Nano R4 pin.
5. **Check the completed board.** Verify every supply and signal connection with a multimeter before inserting the carrier.

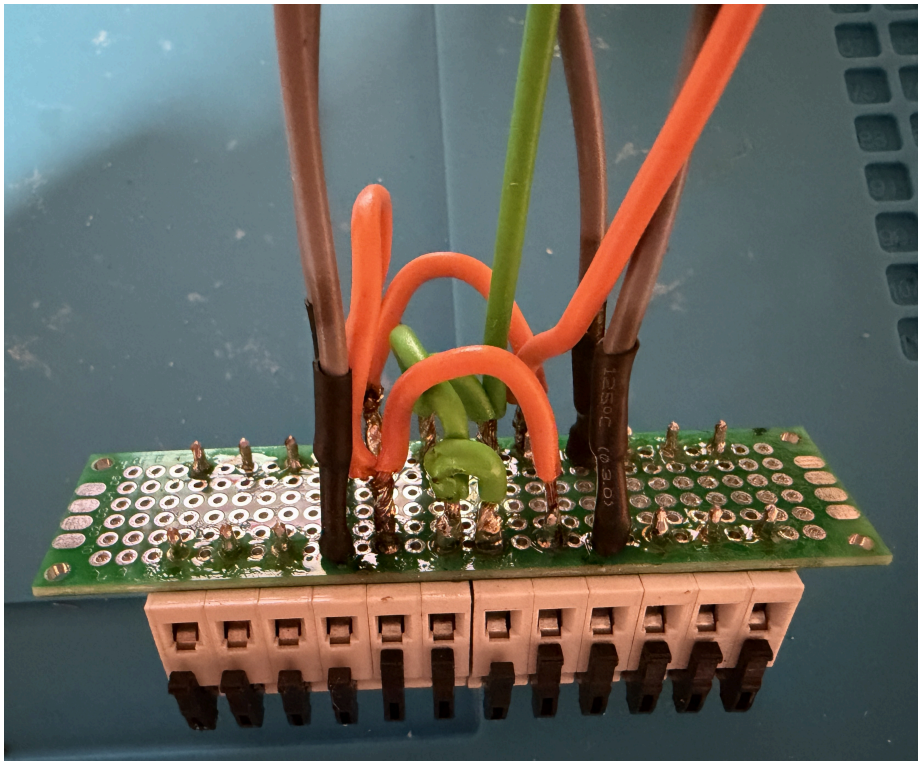


Figure 8: Rear of the sensor carrier with common supply rails and individual signal wires.

8 Arduino Nano R4 wiring

Connect the four signal wires to D2, D3, D8 and D12 according to the schematic. All sensors and the controller need a common ground. The sensor rail comes from the converter; do not route the combined sensor load through an Arduino GPIO pin.

Pin / connection	Purpose
D2, D3, D8, D12	Four independent pulse inputs
GND0, GND1	Common reference for sensors, converter and controller
VIN	Supply from the DC/DC converter; specified VIN range 5–21 V
USB-C	Serial data link; can also supply the board through USB

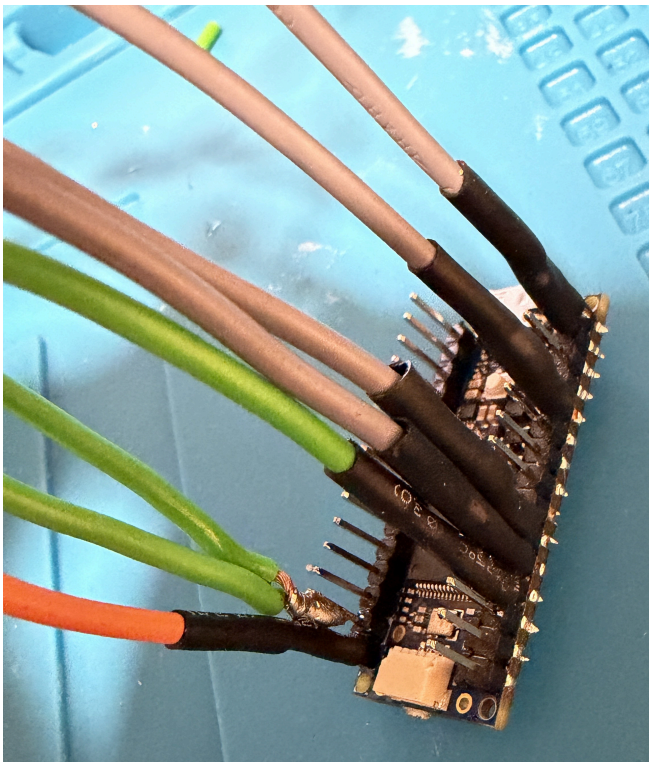


Figure 9: Prototype wiring at the Arduino Nano R4 headers.

The supplied photos document the assembled prototype. Verify pin labels directly on the board and match each input to the channel order used by the firmware.

9 Mechanical assembly and converter setup

1. **Deburr the enclosure.** Check the carrier slots and terminal openings.
2. **Mount the converter.** Keep the adjustment potentiometer and USB connector accessible; prevent conductive parts from contacting unintended wiring.
3. **Connect the converter input.** Observe polarity at the 24 V bus terminal.
4. **Set the supply voltage.** With the sensors disconnected, adjust the converter output to the required supply voltage, verified by a multimeter. Use a setting within the Nano R4′s 5–21 V VIN range and the permitted supply range of every connected sensor. Then switch the supply off.
5. **Connect the Nano R4 supply.** Connect the converter output to VIN and the common ground to GND.
6. **Insert the terminal carriers.** Ensure the enclosure wall fits into the terminal-to-board clearance.
7. **Connect and route the wires.** Keep signal and supply connections secure; avoid pinching wires when the cover is fitted.
8. **Install the cover and latch pin.** Check the DIN-rail attachment and release.

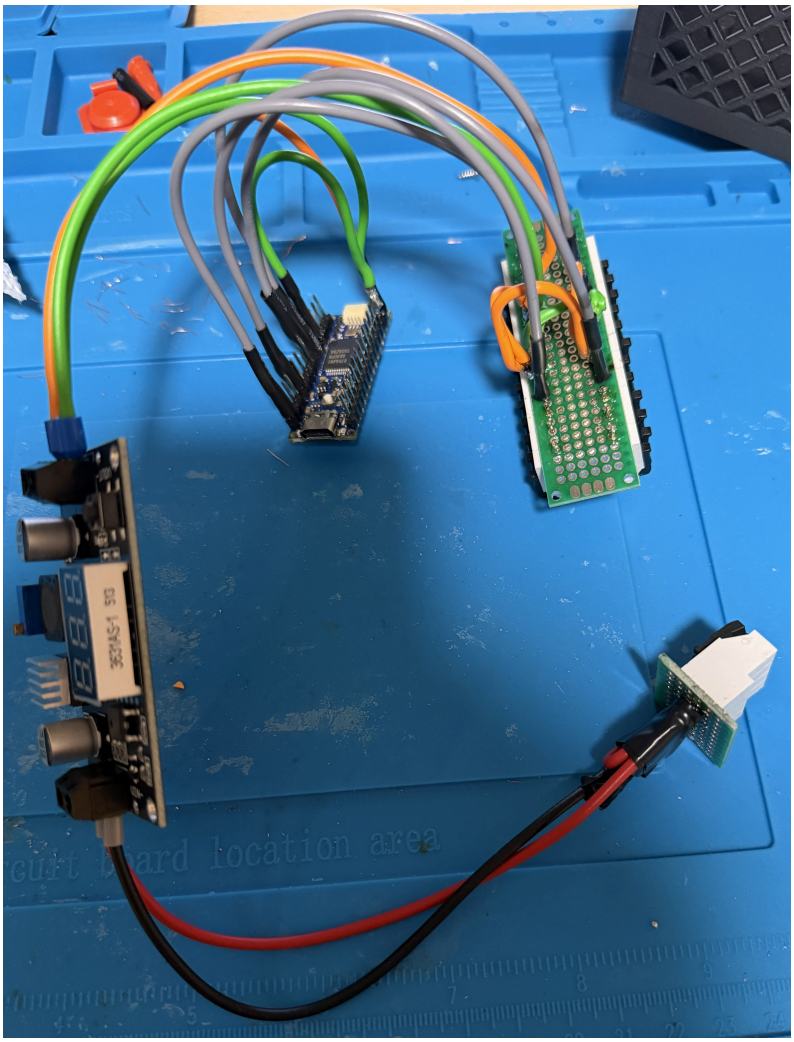


Figure 10: All connections visible without the enclosure.

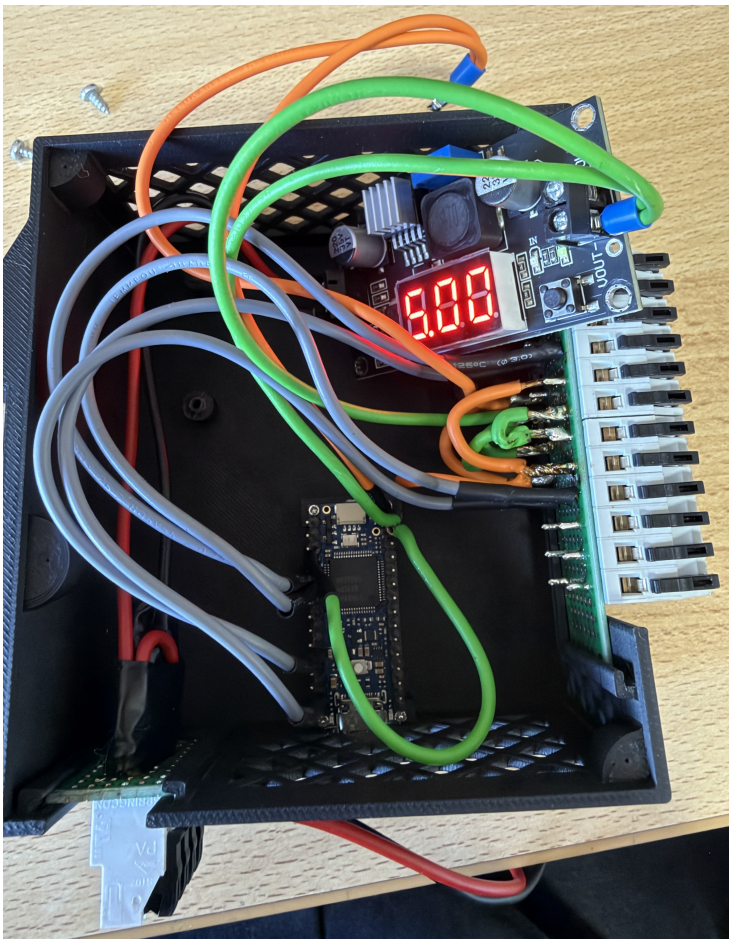


Figure 11: Prototype with the LM2596S display showing approximately 5 V.

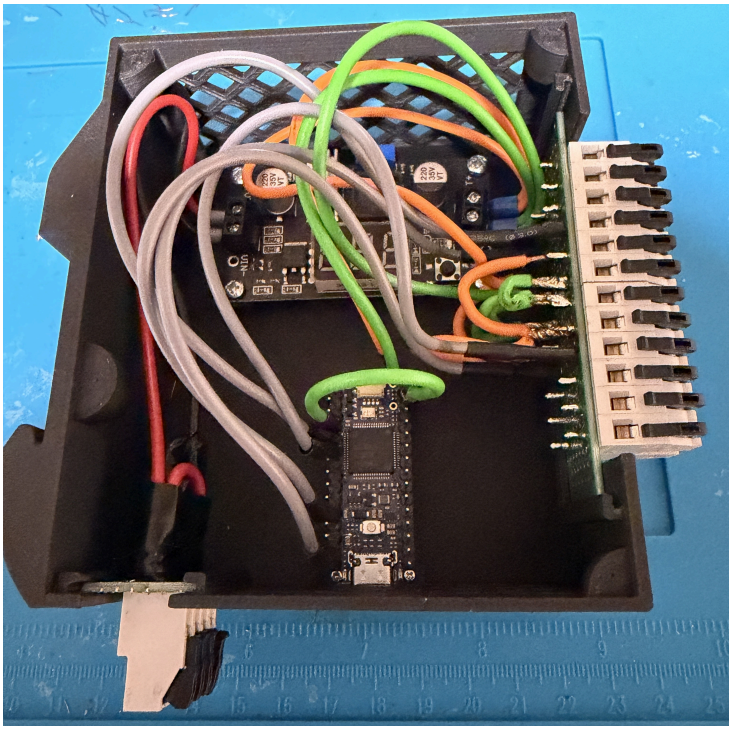


Figure 12: Assembled di-4x-5V prototype with the cover removed.

10 Checks before commissioning

1. **Inspect the wiring and solder joints.** Check polarity, unintended bridges, exposed conductors and strain on the headers.
2. **Check continuity with power off.** Confirm the bus pass-through, common ground, sensor supply connections and the four individual signal paths.
3. **Verify the supply voltage.** Confirm the converter setting is within the permitted supply ranges of the Nano R4 and all connected sensors; maximum 21 V.
4. **Test the converter without sensors.** Use a current-limited supply and confirm the supply rail remains at the selected voltage under the intended load.
5. **Connect one compatible sensor.** Confirm correct supply polarity and that the digital signal remains within the input range.
6. **Test acquisition over USB.** Upload the matching firmware, initialize according to its serial protocol and confirm the assigned channel responds.
7. **Check Pi-side conversion.** For the YF-S201, a received value of 7556 must yield approximately 10.07 l/min after division by 750.
8. **Repeat for all channels.** Match each physical connector to the enclosure label and the corresponding field in the received data.
9. **Check the mechanics.** Verify terminal retention, USB clearance, cover fit and secure DIN-rail mounting.

11 Finished module

The connector labels can be printed from the file `../docs/connector-labels.pdf`.

To attach them to the enclosure, first cover both sides of the printed paper with clear adhesive tape. This helps protect the labels from water and dirt. Then attach the labels to the measurement module using double-sided adhesive tape.

The following three images show where the labels should be placed on the enclosure.



Figure 13: Bottom view with ventilation grid.

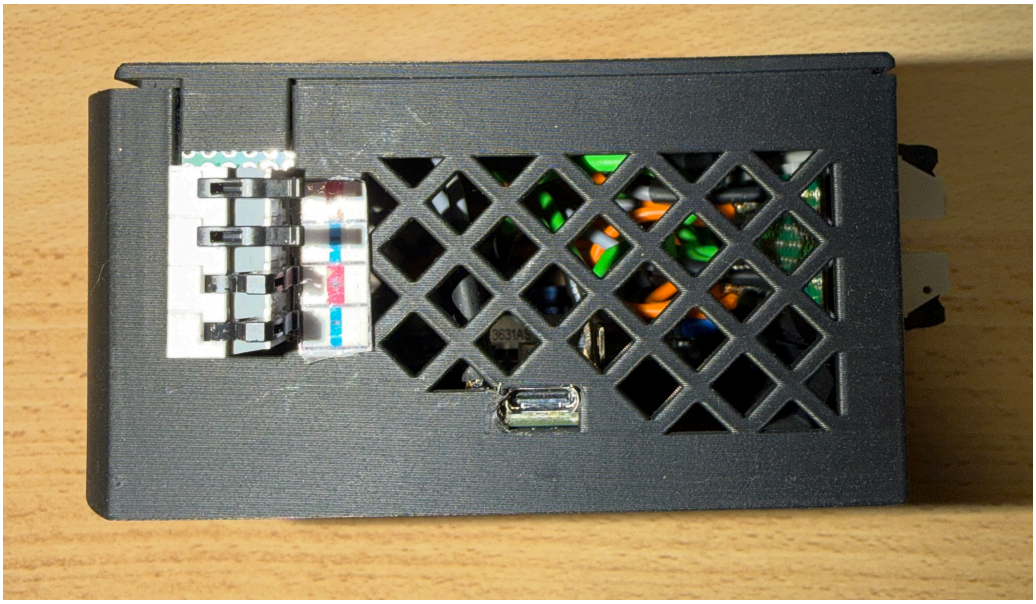


Figure 14: Top view with the 24 V bus terminal and the USB-C socket.

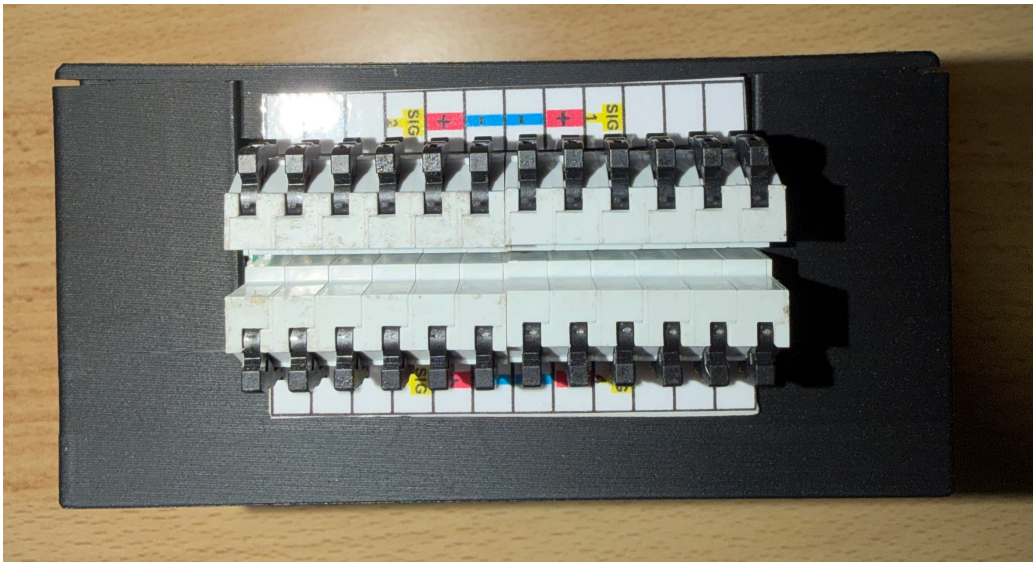


Figure 15: Side view with the signal terminals.